Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/17/2017
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-294-1722
Organization Name:	University of Florida		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	08/26/2011	**End Date:**	07/13/2017
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/13/2017 due to transfer to PI's new institution (Ed., 1/11/2018) NOTE: Extended to 3/31/2019 per NSSC information and PI (Ed., 8/20/2012)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Mulavara, Ajitkumar (Universities Space Research Association) Wood, Scott (NASA Johnson Space Center)		
Grant/Contract No.:	NNX11AR02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NRA NNJ10ZSA003N requested proposals to assess changes in elemental neurocognitive functions such as perception, motor control, memory, attention, language, executive function, and emotional processing following long duration spaceflight using both behavioral assessments and monitoring technologies such as fMRI. In response to this call, we propose to perform structural and functional MR brain imaging to identify the relationship between changes in crewmember neurocognitive function and neural structural alterations following a six month International Space Station mission. Our central hypothesis is that measures of brain structure, function, and network integrity will change from pre to post flight in crewmembers (Aim 1). Moreover, we predict that these changes will correlate with indices of cognitive, sensory, and motor function in a neuroanatomically selective fashion (Aim 2). Our interdisciplinary approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that will be conducted pre flight, during flight, and post flight to investigate neuroplastic and maladaptive brain changes in crewmembers following long duration spaceflight. Success in this endeavor would 1) result in identification of the underlying neural mechanisms and operational risks of spaceflight-induced changes in behavior, and 2) identify whether a return to normative behavioral function following re-adaptation to Earth's gravitational environment is associated with a restitution of brain structure and function or instead is supported by substitution with compensatory brain processes. EDITOR'S NOTE (1/11/2018): Project continues as a new grant, "Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases--80NSSC17K0461," due to Principal Investigator Dr. Seidler's move to University of Florida in July 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this project will have relevance not only to understanding the effects of spaceflight on the human brain and behavior, but also for delineating the capacity of the brain to remodel in response to adaptive stimuli. As such, the results should prove informative for understanding the neural mechanisms associated with adaptive behavioral change and the rehabilitation of these changes during recovery periods.
Task Progress:	We have made substantial progress over the past year, including publication of a paper reporting brain changes with spaceflight from retrospective MRI scans, publication and preparation of the final papers from our bed rest study, and continued data collection on our prospective flight study. Based on the retrospective MRI scans that we acquired, we have published a paper in the journal npgMicrogravity (Koppelmans et al., 2016; see Bibliography section). In this work we evaluated retrospective longitudinal T2-weighted MRI scans and balance data from 27 astronauts (thirteen ~2-week shuttle crew members and fourteen ~6-month International Space Station crew members) to determine spaceflight effects on brain structure, and whether any pre- to postflight brain changes are associated with balance changes. Data were obtained from the NASA Lifetime Surveillance of Astronaut Health. We found extensive volumetric gray matter decreases, including large areas covering the temporal and frontal poles and around the orbits. This effect was larger in International Space Station versus shuttle crew members in some regions. There were bilateral focal gray matter increases within the medial primary somatosensory and motor cortex, i.e., the cerebral areas where the lower limbs are represented. These intriguing findings are observed in a retrospective data set; our ongoing prospective study will help to elucidate the behavioral consequences and longevity of these changes. We are also working with a retrospective diffusion weighted MRI data set. In light of the recent reports of microgravity-induced cephalad fluid shift and gray matter atrophy seen in astronauts, we applied a technique to estimate white matter metrics after correcting for free water contamination. This approach enabled the analysis of white matter tissue-specific alterations that are unrelated to fluid shift, occurring from pre- to post- spaceflight as well as regional shifts in free water (FW). We are in the process of writing up our findings for publication. Prospective longitudinal studies are required to evaluate the time course of recovery from these changes. Data collection and analyses are ongoing for our prospective study. EDITOR'S NOTE (1/11/2018): Project continues as a new grant, "Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases--80NSSC17K0461," due to Principal Investigator Dr. Seidler's move to University of Florida in July 2017.
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